7 Feb 23: Bourgain's Embedding Theorem

Definition 7.1. *Given (X_1, d_1) and (X_2, d_2), a map $\phi : X_1 \to X_2$ has distortion $\alpha \geq 1$ if there exists a scaling parameter $\beta > 0$ such that for all $x, y \in X_1$:*

$$\beta d_1(x, y) \leq d_2(\phi(x), \phi(y)) \leq \alpha\beta d_1(x, y).$$

Theorem 7.2 (Bourgain's Theorem)**.** *Let (X, d) be any n-point metric space. Then there exists an embedding $\phi : (X, d) \to (\mathbb{R}^{O(\log^2 n)}, d_{\ell_1})$ such that the distortion is $O(\log n)$.*

Definition 7.3. *For $i = 1, \ldots, \log n$ and $j = 1, \ldots, c \log n$ (for a sufficiently large constant c), define a random subset $S_{i,j} \subseteq X$ by including each $x \in X$ independently with probability $\frac{1}{2^i}$.*
We define the embedding $\phi : X \to \mathbb{R}^{\log n \times c \log n}$ as:

$$\phi(x) = \big(d(x, S_{1,1}), d(x, S_{1,2}), \ldots, d(x, S_{\log n, c \log n})\big).$$

where $d(x, S) = \min_{s \in S} d(x, s)$.

Theorem 7.4. *With high probability in n, for all $x, y \in X$:*

$$\Omega(\log n) d(x, y) \leq d_{\ell_1}(\phi(x), \phi(y)) \leq c \log^2 n \, d(x, y).$$

This yields an embedding with scaling factor $\beta = \Omega(\log n)$ and distortion $\alpha = O(\log n)$.

Remark 7.5. *Recall that the ℓ_1 distance is the sum of absolute differences across all coordinates. Thus, we need to analyze:*

$$d_{\ell_1}(\phi(x), \phi(y)) = \sum_{i=1}^{\log n} \sum_{j=1}^{c \log n} |d(x, S_{i,j}) - d(y, S_{i,j})|.$$

The upper bound is immediate from the triangle inequality: $|d(x, S) - d(y, S)| \leq d(x, y)$. Summing this over all $c \log^2 n$ coordinates gives the upper bound $c \log^2 n \, d(x, y)$. The core of the proof is establishing the lower bound.

Proof. Fix $x, y \in X$. For any $z \in X$ and $r \geq 0$, let $B(z, r) = \{w \in X \mid d(z, w) \leq r\}$ be the closed ball, and let $B^\circ(z, r) = \{w \in X \mid d(z, w) < r\}$ be the open ball.
For each i, define $\rho_i(x) = \inf\{r \mid |B(x, r)| \geq 2^i\}$. By definition, the open ball is strictly smaller than 2^i, so $|B^\circ(x, \rho_i(x))| < 2^i$, while the closed ball satisfies $|B(x, \rho_i(x))| \geq 2^i$.
Define $r_i = \max(\rho_i(x), \rho_i(y))$. Without loss of generality, assume $r_i = \rho_i(x)$. This gives us two critical observations:

1. $|B^\circ(x, r_i)| < 2^i$ (since $r_i = \rho_i(x)$).
2. $|B(y, r_i)| \geq 2^i$ (since $r_i \geq \rho_i(y)$). Since balls only grow with radius, it also holds that $|B(y, r_{i-1})| \geq 2^{i-1}$.

Define t as the largest integer such that $r_t \leq \frac{d(x,y)}{3}$. Define $r_0 = 0$, and cap the sequence by defining $r_{t+1} = \frac{d(x,y)}{3}$.

Lemma 7.6. *For any $i \in \{1, \ldots, t+1\}$ and $j \in \{1, \ldots, c \log n\}$, with constant probability $p > 0$ (specifically $p > \frac{1}{10}$),*

$$|d(x, S_{i,j}) - d(y, S_{i,j})| \geq r_i - r_{i-1}. \qquad (*)$$

Proof of Lemma: To satisfy $(*)$, it is sufficient for $S_{i,j}$ to miss the open ball $B^\circ(x, r_i)$ and hit the closed ball $B(y, r_{i-1})$. Since $r_i \leq \frac{d(x,y)}{3}$ and $r_{i-1} \leq \frac{d(x,y)}{3}$, the distance between the centers of these two balls is $d(x, y) \geq r_i + r_{i-1} + \frac{1}{3} d(x, y)$. Thus, $B^\circ(x, r_i)$ and $B(y, r_{i-1})$ are completely disjoint, making hitting and missing them completely independent events.

- **Miss Event:** $\Pr[S_{i,j} \cap B^\circ(x, r_i) = \emptyset] = \left(1 - \frac{1}{2^i}\right)^{|B^\circ(x,r_i)|} \geq \left(1 - \frac{1}{2^i}\right)^{2^i} \approx \frac{1}{e}$.
- **Hit Event:** $\Pr[S_{i,j} \cap B(y, r_{i-1}) \neq \emptyset] = 1 - \left(1 - \frac{1}{2^i}\right)^{|B(y,r_{i-1})|} \geq 1 - \left(1 - \frac{1}{2^i}\right)^{2^{i-1}} \approx 1 - \frac{1}{\sqrt{e}}$.

By independence, the probability that both occur is at least $\frac{1}{e}\left(1 - \frac{1}{\sqrt{e}}\right) \approx 0.14 > \frac{1}{10}$. When this happens, $d(x, S_{i,j}) \geq r_i$ and $d(y, S_{i,j}) \leq r_{i-1}$, so the absolute difference is at least $r_i - r_{i-1}$.

Claim 7.7. *For any $i \leq t+1$, let Z_i be the number of indices j such that $(*)$ holds. Then $Z_i \geq \Omega(\log n)$ with high probability.*

Proof of Claim: Z_i is the sum of $c \log n$ independent 0-1 random variables, each with mean $\geq \frac{1}{10}$. The expected value is $\mu \geq \frac{c \log n}{10}$. By the Chernoff bound, for a sufficiently large constant c, the probability that $Z_i < \frac{c \log n}{20}$ is exponentially small ($o(1/n^3)$). Applying a union bound over all pairs x, y and all i, this bound holds uniformly with high probability. Let $K = \frac{c}{20}$, so $Z_i \geq K \log n$.

Using the claim, we can now telescope the sum for the lower bound:

$$\begin{aligned}
d_{\ell_1}(\phi(x), \phi(y)) &= \sum_{i=1}^{\log n} \sum_{j=1}^{c \log n} |d(x, S_{i,j}) - d(y, S_{i,j})| \\
&\geq \sum_{i=1}^{t+1} \sum_{j:(*) \text{ holds}} (r_i - r_{i-1}) \\
&\geq \sum_{i=1}^{t+1} (K \log n)(r_i - r_{i-1}) \\
&= K \log n \sum_{i=1}^{t+1} (r_i - r_{i-1}) \\
&= K \log n (r_{t+1} - r_0) \\
&= K \log n \frac{d(x, y)}{3} \\
&= \Omega(\log n) d(x, y).
\end{aligned}$$

This completes the proof of the lower bound, showing the overall distortion is bounded by $O(\log n)$. $\square$

8 March 10: Metric Embedding Day 2

Definition 8.1 (Tree Metric)**.** *Let (V, d) be a metric space. A tree metric is defined by a tree $T = (V_T, E_T)$ with non-negative edge weights, such that $V \subseteq V_T$. The tree distance $d_T(x, y)$ between any two nodes $x, y \in V$ is the sum of the edge weights along the unique simple path connecting them in T.*
We say T embeds (V, d) with distortion $\alpha \geq 1$ if for all $x, y \in V$:

$$d(x, y) \leq d_T(x, y) \leq \alpha d(x, y).$$

Theorem 8.2. *Let (X, d) be n point space. Suppose $\forall x \neq y \in X$, $d(x, y) \in [1, n^2]$. We will embed X into a tree metric (T, d_T) with*
1. *depth of tree is $O(\log n)$.*
2. *For any root-leaf path, edges weights are decreasing powers of 2.*
3. *Points in X are mapped to leaves of T*

There exist randomized embedding $\phi : X \to (T, d_T)$ such that $\forall x, y \in X$, $d_T(\phi(x), \phi(y)) \geq d(x, y)$ and $\mathbb{E}[d_T(\phi(x), \phi(y))] \leq O(\log n) d(x, y)$

Proof. The algorithm is as follows:

1. Pick random permutation of elements of X say $\pi = \pi_1, ..., \pi_n$
2. Pick uniformly random $\beta \in [1, 2)$ and define $\beta_i := 2^{i-1}\beta$ and for each $x \in X$ and $i = 0, ..., 2 \log n$ and define $center_i(x) :=$ first π_j to satisfy $d(x, \pi_j) \leq \beta_i$
3. Construct as follows: Each node represents sets of elements with root at level $2 \log n + 1 =$ all of X. For each node w at level $i + 1$, if node > 1 element, group w's elements z by $center_i(z)$. Edges from level $i + 1$ to level i has weight $2^i \beta$

Analysis:
We claim: For $x, y \in X$, $d_T(\phi(x), \phi(y)) \geq d(x, y)$. To prove the claim, observe at level i, if x, y in same node then $d(x, y) \leq 2\beta_i \leq 2^{i+1}$. Let $i =$ highest level path goes through between x and y. $d_T(\phi(x), \phi(y)) \geq 2 \cdot 2^i \geq d(x, y)$.
Property: $\mathbb{E}[d_T(\phi(x), \phi(y))] \leq O(\log n) d(x, y)$
To prove this property, $x := \phi(x), y := \phi(y)$,we can trivially say $d_T(x, y) \leq O(\text{largest edge weight along path})$. This is upper bounded by $\sum_{i=0}^{2 \log n} 2^{i+1} \mathbb{I}[center_i(x) \neq center_i(y)]$. Taking the expectation we get

$$\begin{aligned}
\mathbb{E}[d_T(x, y)] &\leq \sum_{i=0}^{2 \log n} \int_{r=2^{i-1}}^{2^i} \Pr[B_i = r] \Pr[center_i(x) \neq center_i(y) | B_i = r] \cdot 2^{i-1} dr \\
&= \sum_{i=0}^{2 \log n} \int_{r=2^{i-1}}^{2^i} \frac{1}{2^{i-1}} \Pr[center_i(x) \neq center_i(y) | B_i = r] \cdot 2^{i-1} dr \\
&= \sum_{i=0}^{2 \log n} \Pr[center_i(x) \neq center_i(y) | B_i = r] dr \qquad (*)
\end{aligned}$$

Say that $w \in X$ 'cuts' (x, y) at level i if we look through $\pi_1, ...,$ the first of $\{center_i(x), center_i(y)\}$ to appear is w implies w cuts (x, y) and the elements are non-equal.
Let $w_1, w_2, ..., w_n$ be elements of X sorted by distance from w_i to $\{x, y\}$.

$$* \leq \sum_{j=1}^{n} \sum_{i=0}^{2 \log n} \int_{r=2^{i-1}}^{2^i} \Pr[w_j \text{ cuts } (x, y) \text{ at level } i | B_i = r] dr.$$

For w_j to cut (x, y) suppose wlog $d(v_j, x) \leq d(v_j, y)$ we need

1. $d(w_j, x) \leq \beta_i \leq d(w_j, y)$
2. w_j has to appear before $w_1, ..., w_{j-1}$ and this occurs with probability $\frac{1}{j}$

So we can bound the probability as

$$\leq \sum_{j=1}^{n} \int_{d(w_j,x)}^{d(w_j,y)} \frac{1}{j} dz = \sum_{j=1}^{n} \frac{1}{j} \left(d(w_j,y) - d(w_j,x)\right) \leq d(x,y) O(\log n).$$

□

9 March 16

Theorem 9.1. *There exist history independent algorithm with expected cost* $\tilde{O}\left(\log^{1.5} n\right)$.

Theorem 9.2. *There exist history independent algorithm such that if instead is to a uniform random rank, then expected insertion cost is* $\tilde{O}(\log^{1.5} n)$.

Remark 9.3. *One potential approach that is incorrect is instead of candidates being median* $\theta(\frac{t}{\log n})$ *elements in subproblem we choose it to be median* $\frac{t}{100\sqrt{\log n \log\log n}}$ *elements in subproblem.*
The good news is that

$$\Pr[\textit{insertion into subprob changes pivot}] = O\left(\frac{1}{|Cand|}\right).$$

This implies the expected cost of insertion is $\tilde{O}\left(\log^{1.5} n\right)$.
However the algorithm is not an algorithm as density could be as large as

$$\frac{1}{2} + \theta(\log n) \cdot \theta\left(\frac{\tilde{1}}{\sqrt{\log n}}\right) >> 1.$$

Proof. The fix is that if density(subproblem)$\notin [\frac{1}{4}, \frac{3}{4}]$ revert back to using the $O(\log^2 n)$ solution on this subproblem and all of its descendents. Call this subproblem and its descendents 'corrupt nodes'.
Note: This needs all densities between say $\frac{1}{8}$ and $\frac{7}{8}$. □

Lemma 9.4. *Fraction of the leaves that are corrupt is* $\leq \frac{1}{\log^1 0n}$

Proof. Let $A_1, A_2, ...$ be the number of paths to the leaf. Let $\triangle_i = density(A_{i+1}) - density(A_i)$.

$$density(A_j) = \frac{1}{2} + \triangle_1 + \cdots + \triangle_{j-1}.$$

We want to show that

$$\left|\sum_{i=1}^{j-1} \triangle_i\right| < \frac{1}{4}.$$

We have that $|\triangle_i| \leq \frac{1}{100\sqrt{\log n \log\log n}}$ and $\mathbb{E}[\triangle_i | \triangle_1, ..., \triangle_{i-1}] = 0$.
Suppose $\triangle_1, \triangle_2, ..., \triangle_\ell \in [-s, s]$ and $\mathbb{E}[\sum_{i=1}^{\ell} \triangle_i] = 0$ then

$$\Pr[|\sum_{i=1}^{\ell} \triangle_i| > K\sqrt{\ell} \cdot S] \leq 2e^{-\frac{K^2}{4}}.$$

For us. $\ell \leq \log n$, $S = \frac{1}{100\sqrt{\log n \log\log n}}$ and set $K = 10\sqrt{\log\log n}$ then

$$2e^{-\frac{K^2}{4}} = 2e^{-2S\log\log n} \approx \frac{1}{(\log n)^{25}} \text{ and } K\sqrt{\ell}S = \frac{1}{10}.$$

So

$$\Pr\left[\left|\sum_{i=1}^{j} \triangle_i\right| \geq \frac{1}{10}\right] \leq \frac{1}{(\log n)^{25}}.$$

So the $\Pr[i\text{-th node } A_i \text{ makes us corrupt}] \leq \frac{1}{(\log n)^{25}}$. By union bound,

$$\Pr[\text{corrupt}] \leq \log n \frac{1}{(\log n)^{25}} \leq \frac{1}{(\log n)^{10}}.$$

□

Theorem 9.5. *Consider insertion to random rank with fraction of leaves corrupt being* $\frac{1}{\log^{10} n}$. *Additionally, levels have densities in* $[\frac{1}{8}, \frac{7}{8}]$ *implies*

$$\Pr[\textit{insertion is to corrupt leaf}] = O\left(\frac{1}{\log^1 0n}\right).$$

If insertion does not hit corrupt node: $\mathbb{E}[cost] = \tilde{O}(\log^{1.5} n)$.
If insertion hits correct nodes: $\mathbb{E}[cost] = O(\log^2 n)$ *and happens with probability* $\frac{1}{\log^1 0n}$.

Proof. Consider arbitrary insertion, we can use a modified algorithm

1. Use array of size $4n$
2. Select random $q \in [n]$
3. Add dummy elements $<$ all real elements
4. Add $n - q$ dummy elements $>$ all real elements

The rank including dummy elements is $q + r \in [r, n + r]$.

$$\mathbb{E}[cost] = \sum_{i=r}^{n+r} \mathbb{E}[\text{cost of ins rank } i] O\left(\frac{1}{n}\right) \leq \sum_{i=1}^{2n} O\left(\frac{1}{n}\right) \mathbb{E}[\text{cost of ins at rank } i] = \tilde{O}(\log^{1.5} n).$$

□

10 March 18 — Optional Stopping Theorem

Definition 10.1. *Consider sequence of random variables $X_0, X_1, ..., X_S$ (not random)*

1. *X_0 is amount you start with*
2. *X_i = amount you have after i steps*
3. *S is the stopping time and is a random variable determined by $X_0, ..., X_s$ and is finite with probability 1.*

We require for any $i > 0$

$$\mathbb{E}[X_i | X_0, X_1, ..., X_{i-1}, S > i-1] = X_{i-1}.$$

Theorem 10.2. *Optional Stopping Theorem: If $\mathbb{E}[S]$ finite and $\exists C$ such that for all $i, \mathbb{E}[|X_i - X_{i-1}||S > i-1] \leq C$. Then $\mathbb{E}[X_S] = X_0$*

Proof. Define $\Delta_i = X_i - X_{i-1}$, wewant to show that $\mathbb{E}[\sum \delta_i] = 0$. Define $\Delta_i^+ = \begin{cases} \Delta_i & \text{if } \Delta_i \geq 0 \\ 0 & \text{if } \Delta_i < 0 \end{cases}$ and $\Delta_i^- = \begin{cases} \Delta_i & \text{if } \Delta_i < 0 \\ 0 & \text{if } \Delta_i \geq 0 \end{cases}$ and note that $\Delta_i = \Delta_i^+ - \Delta_i^-$.

$$\mathbb{E}\left[\sum \Delta_i^+\right] = \sum \mathbb{E}\left[\Delta_i^+\right] \leq C \sum \Pr[S \geq i] = C\mathbb{E}[S].$$

Likewise, $\mathbb{E}\left[\sum \Delta_i^-\right]$ is finite.

$$\mathbb{E}[\sum \Delta_i] = \mathbb{E}\left[\sum \Delta_i^+ - \sum \Delta_i^-\right] = \mathbb{E}\left[\sum \Delta_i^+\right] - \mathbb{E}\left[\sum \Delta_i^-\right] = 0.$$

□

Example 10.3. *Consider the game*

1. *Start at n*
2. *On each step, go left with prob $\frac{2}{3}$ and right with prob $\frac{1}{3}$*
3. *Stop when you get to 0*

Let $S = \#$ steps until you get to 0. What is $\mathbb{E}[S]$.
Let P_i = position at step i and define $X_i = P_i + \frac{1}{3}i$ then

$$\mathbb{E}[X_i | ...] = X_{i-1} + 1 \cdot \frac{1}{3} + (-1) \cdot \frac{2}{3} + \frac{1}{3} = X_{i-1}.$$

We have $n = X_0 = \mathbb{E}[X_S] = \mathbb{E}\left[P_S + \frac{S}{3}\right] = \frac{1}{3}\mathbb{E}[S]$ implying $\mathbb{E}[S] = 3n$.

11 March 23

Example 11.1. *Start at* 1 *and perform a random walk where* $+1$ *with probability* $\frac{2}{3}$ *and* -1 *with probability* $\frac{1}{3}$.
What is $\Pr[$*get to* n *before* $0]$*?*

Proof. Define $X_t =$ position after t steps and $Y_t = \frac{1}{2^{X_t}}$.
Observe that $\mathbb{E}[Y_{t+1}|Y_t] = Y_t$.
Let Y_S be the point where we hit 0 or n. By optional stopping theorem, $\mathbb{E}[Y_S] = Y_0 = \frac{1}{2}$ and

$$\mathbb{E}[Y_S] = p \cdot \frac{1}{2^n} + (1-p) \cdot 1.$$

So $p = \frac{1}{2(1-\frac{1}{2^n})}$ □

12 March 25

12.1 Linear probing hash table — setup

We store a set of at most n keys drawn from a universe $\mathcal{U}$ in a table T of m slots indexed by $\{0, 1, \ldots, m-1\}$. Each slot holds at most one key (or is empty). We use a hash function $h\colon \mathcal{U} \to \{0, \ldots, m-1\}$; in the randomized model, h is drawn from a suitable family (e.g. a universal or strongly universal family), or we analyze a fully random function h.

Insertion. To insert key k, examine slots

$$h(k),\ h(k)+1,\ h(k)+2,\ \ldots \pmod{m}$$

in order until finding an empty slot, and place k there.

Search. To search for k, probe the same sequence starting at $h(k)$ until either finding k or hitting an empty slot (then k is absent).

Load. Write $\alpha = n/m$ for the load factor. Analysis usually assumes α is bounded away from 1 (often $\alpha \le 1 - \varepsilon$ for constant $\varepsilon > 0$) so that the table is not full and the walk terminates.

Clusters. Keys occupying consecutive occupied slots form *runs* or *clusters*. Linear probing tends to lengthen clusters on collision; the main analytic quantity is the length of the probe sequence (number of slots examined) for an operation.